Code No. 3527

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) V-Semester (MAIN) Examinations Dec 2023 /Jan 2024

DATA MINING

Time : 3 hours **Max. Marks : 70**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7).*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	How does the choice of a similarity measure impact the clustering results in data mining.	2	1	2
b)	Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8), Compute the supremum distance between the two objects.	2	3	1
c)	What is the significance of the confidence measure in association rule mining.	2	1	3
d)	State Bayes' theorem.	2	1	3
e)	What is pruning in the context of decision trees.	2	1	3
f)	Depict the role of a dendrogram in hierarchical clustering.	2	2	1
g)	Name two types of ensemble methods.	2	1	3
2. a)	Discuss various techniques of Data Visualization.	7	4	3
b)	Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Calculate the standard deviation of the age values.	7	5	2
3. a)	Illustrate a comparison of Pattern Evaluation Measures.	7	2	3
b)	A database has five transactions. Let min sup = 60% and min conf = 80%.	7	5	1

TID	*Items bought*
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y }
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

Find all frequent itemsets using FP growth.

(P.T.O.)

4.	a)	Describe classification by back propagation.	7	3	3
	b)	Illustrate a Multilayer Feed-Forward Neural Network.	7	2	3
5.	a)	Differentiate between Density-Based Methods and Grid-Based Methods.	7	4	2
	b)	Explain the concept of core points, border points, and noise points in DBSCAN.	7	1	3
6.	a)	Explain the process of association rule mining with an example.	7	1	3
	b)	Discuss the balance between using data mining for public good and individual privacy.	7	4	3
7.	a)	Discuss ensemble learning methods.	7	4	3
	b)	Why is naive Bayesian classification called "naive". Briefly outline the major ideas of naive Bayesian classification.	7	1	2

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